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Mangroves and their services are at risk from tropical cyclones and sea level rise under climate change

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Abstract
Climate change is expected to alter the frequency and intensity of extreme events, modifying the natural disturbance regimes to which ecosystems are currently adapted. Here, we present a spatially explicit risk index for mangroves and their associated biodiversity and ecosystem services based on projected frequency changes of tropical cyclone wind speeds and rates of relative sea level rise under SSPs 245, 370 and 585 by 2100. Globally, approximately half of the total mangrove area (40-56% depending on the SSP) will be at high to severe levels of risk due to climate-modified tropical cyclone disturbance regimes. Further, we find mangrove areas with high levels of biodiversity and ecosystem services provision, including coastal protection for people and assets, carbon sequestration, and fishery benefits, are at proportionally higher levels of risk than mangrove forests generally. Our findings emphasize the need to anticipate changes in natural disturbance regimes and to adapt ecosystem management to sustain ecosystem services in the future.

Main
Mangrove ecosystems provide vital contributions to human well-being, that are increasingly threatened by climate change. The mangrove ecosystem services of carbon sequestration and disaster risk reduction are especially important for climate change mitigation and adaptation respectively (Menéndez et al., 2020; Song et al., 2023). However, these ecosystems are subject to both slow-onset climatic changes, such as sea-level rise, and changes in the intensity and frequency of extreme events, such as tropical cyclones. Both can have potentially large impacts on ecosystem service provision by mangroves; however, threats to mangroves from tropical cyclone frequency changes and sea level rise have been studied separately (Kropf et al., 2024; Mo et al., 2023; Saintilan et al., 2023; Schuerch et al., 2018), and often without considering ecosystem services, or restoration and conservation priorities (Dabalà et al., 2023; T. Worthington & Spalding, 2018). Thus, an assessment of risk that accounts for changes in ecosystem disturbance through tropical cyclones and simultaneous sea level rise for both mangroves and their services is urgently needed.

Mangroves in the world's tropical cyclone basins are adapted to their historical disturbance regimes (Krauss & Osland, 2020; Mo et al., 2023), yet novel disturbance regimes induced by

climate change, with greater frequency and intensity of extreme events than historically, can substantially alter ecosystem composition, structure, and function (Johnstone et al., 2016; Turner & Seidl, 2023). Investigating tropical cyclone risk to mangroves under climate change., Mo et al. (2023) project substantial increases in tropical cyclone frequency at the country-level in North and Central America under 2°C warming, and Kropf et al. (2024) found widespread risk for most mangrove ecoregions affected by tropical cyclones under SSP585 until 2050. Tropical cyclone frequency, alongside temperature and precipitation, plays a critical role in limiting mangrove canopy height (Simard et al., 2019) and affects forest composition and functional traits (Hogan et al., 2018), which raises the concern that climate-driven changes in tropical cyclones could lead to significantly altered mangrove ecosystems. Further, increased tropical cyclone frequency could prevent mangroves from recovering fully in between events, potentially decreasing mangrove area (Sippo et al., 2018) and, in extreme cases, inducing ecological regime shifts by converting mangrove forest to mudflats (Osland et al., 2020). These potential changes in mangrove forest structure and function could alter the biodiversity that depends on these systems, ecosystem services (e.g., carbon sequestration, coastal protection, fisheries), and the role of mangroves in climate mitigation and adaptation.

Under climate change, the impact of tropical cyclones on mangroves is further compounded by sea level rise. Biogenic feedbacks can allow mangroves to adjust vertically to sea level rise, however, high rates of sea-level rise may lead to elevation deficits beyond mangrove adaptive capacity (Saintilan et al., 2023; Schuerch et al., 2018). Saintilan et al. (2023) project widespread loss of coastal ecosystems, including mangroves, due to rates of sea level rise exceeding adaptive capacity at warming levels of 1.5-4.0°C. While sea level rise constitutes a chronic stressor and tropical cyclones are an acute disturbance, both can cause mangrove mortality and subsequently peat collapse (Chambers et al., 2019). Thus, the capacity of mangroves to adapt to sea level rise may be compromised by tropical cyclone-induced mortality, and vice versa, leading to peat collapse and surface elevation loss (Osland et al., 2020). Hence, interactions with sea level rise play a key role in the risks of a changing tropical cyclone disturbance regime, necessitating studies that consider both.

This combined risk from changes in tropical cyclones and sea level rise is relevant to conservation and restoration planning in ways that have not been fully considered to date. Mangroves play a critical role in climate mitigation and adaptation (Lovelock & Reef, 2020; Menéndez et al., 2020), and millions of people in coastal areas around the world depend on mangroves for coastal protection, nutrition and income (Selig et al., 2019; zu Ermgassen et al., 2021). Consequently, mangrove protection and restoration are receiving increasing attention from conservation actors and policy (Spalding & Leal, 2024; T. Worthington & Spalding, 2018). While these initiatives recognize the relevance of sea level rise and altered tropical cyclones under climate change, they do not include comprehensive future risk assessments for the mangrove ecosystem services these initiatives desire to increase. Structural changes, degradation or loss of mangrove forests triggered by a novel disturbance regime could have cascading impacts, resulting in the loss of the vital services provided by these ecosystems. For instance, storm-induced damages can compromise the protective function of mangrove forest for future events and increase risk for coastal communities and assets (Krauss & Osland, 2020). Mangrove carbon stocks can be negatively affected by post-storm disturbance processes such

as peat collapse (Lovelock & Reef, 2020). Since mangroves serve as habitats for many fish and invertebrate species, fishery benefits are threatened by mangrove loss (Selig et al., 2019; zu Ermgassen et al., 2021). To support ecosystem management decisions adaptive to climate change, understanding potential losses of ecosystems services induced by novel disturbance regimes is crucial (Leverkus et al., 2021; Turner & Seidl, 2023).

Here, we assess the risk of a disturbance regime shift for mangroves (Bunting et al., 2018) under climate-modified tropical cyclone activity (Emanuel, 2008; Emanuel et al., 2006) and sea level rise (Garner et al., 2021; Saintilan et al., 2023). We consider a disturbance regime to be "the spatial and temporal dynamics of disturbance across a landscape over a long time" (Turner & Seidl, 2023). In this context, we define risk as a combination of shifts in the frequency with which mangroves experience a given tropical cyclone windspeed intensity, and the rate of relative sea level rise until the end of the century. We combine our spatially explicit risk index with the global mangrove conservation priority areas by Dabalà et al. (2023), which rank mangrove area based on biodiversity, coastal protection for people and property, carbon sequestration and fishery benefits. We analyze where and to what extent the ecosystem services of coastal protection for people and property (Menéndez et al., 2020), fishery benefits (zu Ermgassen et al., 2021), and carbon sequestration (Saintilan et al., 2023; Sanderman et al., 2018) are at risk. Across SSPs, we find about half of global mangrove extents at risk by the end of the century, with areas of high ecosystem service provision at proportionally higher levels of risk.

Changes in tropical cyclone disturbance for global mangrove extents

The IPCC defines risk as a combination of hazard, exposure, and vulnerability (Pörtner et al., 2022). Since mangroves are adapted to their historical disturbance regime (Krauss & Osland, 2020), we do not consider them vulnerable to tropical cyclone hazards under baseline conditions. However, novel disturbance regimes are characterized by a departure from baseline conditions (Turner & Seidl, 2023). We therefore define tropical cyclone disturbance regime shift risk (from here on "risk") for mangroves (**Figure 1**, Panel A), as a combination of shifts in the frequency with which mangroves experience a given tropical cyclone windspeed intensity (Figure 1, Panel B), and the rate of relative sea level rise (Figure 1, Panel C) until the end of the century. Based on the scientific literature (Osland et al., 2020), we assume sea level rise to increase vulnerability to tropical cyclones, and *vice versa*, which we translate into in the five risk categories (low, medium, high, very high, severe) reflected in the risk matrix (**Figure 1**, Panel A). A low to medium risk rank reflects rates of sea level rise mangroves are likely able to adapt to in combination with an increased frequency of low tropical cyclone windspeed intensities. A high or very high risk rank reflects a combination of high to very high rates of sea level rise combined with changes in high tropical cyclone windspeeds. Finally, a severe risk rank indicates a combination of sea level rise rates that mangroves are highly unlikely to adapt to with increased frequency in very high windspeeds.

The probabilistic tropical cyclone data used in this study is downscaled from a statistical-dynamical tropical cyclone model (Emanuel et al., 2006; 2008), which is forced by CMIP6 global climate model outputs across various SSPs for the present day and future climate. The tropical cyclone simulations over the historical period establish a baseline that realistically characterizes

the current tropical cyclone disturbance regime, overcoming the limitations of the historical record of observations (Meiler et al., 2022). Meanwhile, future tropical cyclone data represents future disturbance regimes by the end of the century. Our tropical cyclone windspeed intensity categories correspond to 1) 33-49 m s^{-1}, 2) 50-70 m s^{-1}, 3) >70 m s^{-1}, which are equivalent to Saffir-Simpson Hurricane Wind Scale categories 1 to 2, 3 to 4, and 5 respectively (National Hurricane Center, 2022). We consider mangroves to experience a novel disturbance regime if they are newly affected by windspeed intensities of a given intensity category, or if the relative change in frequency within the category exceeds 100% (**Figure 1**, Panel B). Only windspeed intensities with return periods of up to 250 years are considered ecologically relevant and included (Kropf et al., 2024).

We define our sea level rise categories as 1) 0-4 mm y^{-1}, 2) 4-7 mm y^{-1}, and 3) 7 mm y^{-1} of median regional rate of sea level rise until the end of the century (**Figure 1**, Panel C). Previous research has investigated potential thresholds in the vertical adjustment capacity of mangroves, and found deficits are likely at sea level rise rates of 4 mm y^{-1}, and highly likely at 7 mm y^{-1} (Saintilan et al., 2023).

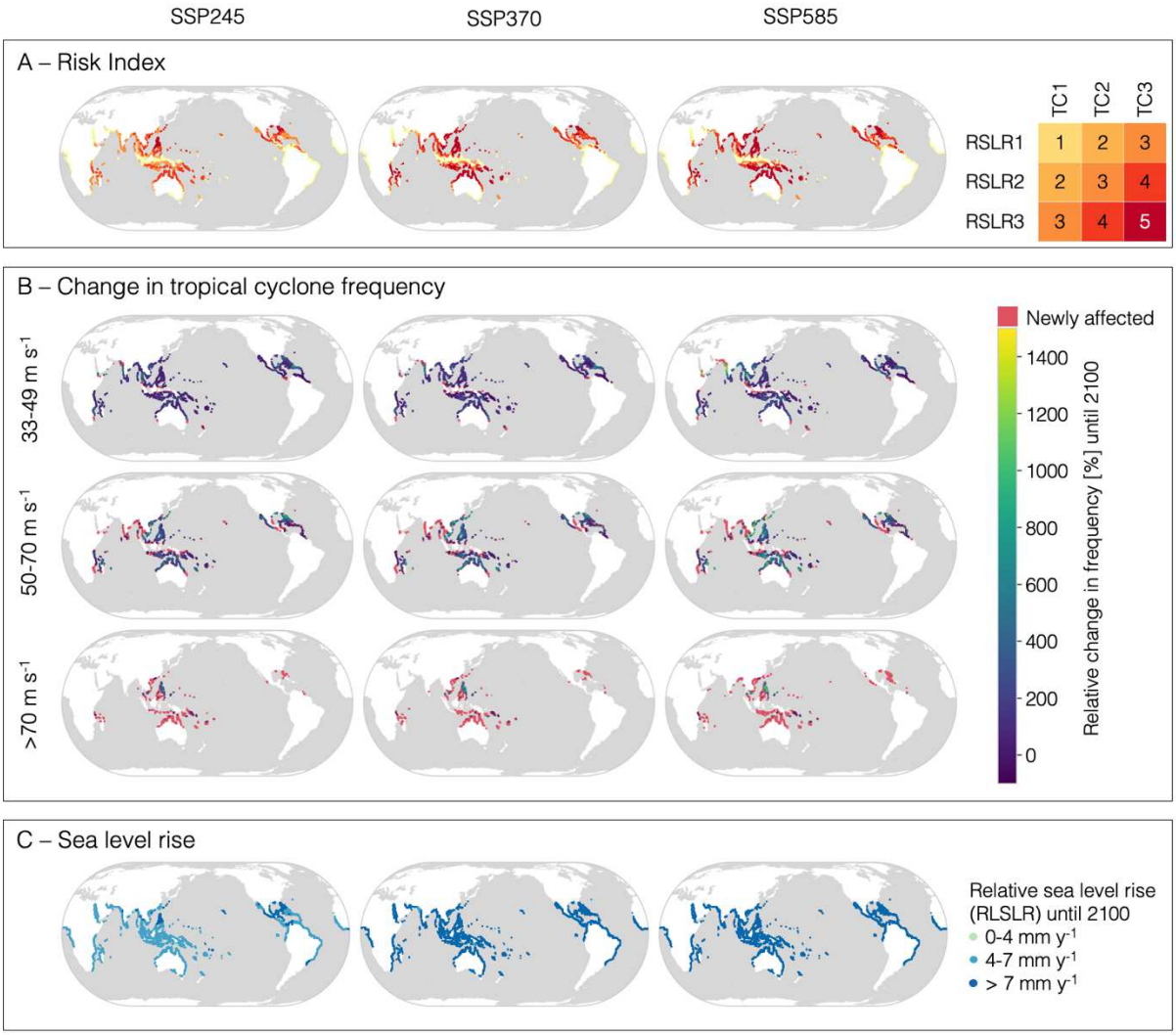

Figure 1: Composite tropical cyclone and sea level rise risk index for mangroves. *A) Risk index at global mangrove locations for SSP245, 370, and 585 until 2100. The legend is the risk matrix showing relative sea level rise (RSLR) and tropical cyclone (TC) risk factors and the resulting risk rank. Risk rank 1-5 correspond to low, medium, high, very high, and severe risk. RSLR 1-3 corresponds to the median rates of regional relative sea level rise of 0-4 mm y^{-1}, 4-7 mm y^{-1}, and 7 mm y^{-1}, while TC1-3 refers to the windspeed categories 1) 33-49 m s^{-1}, 2) 50-70 m s^{-1}, 3) >70 m s^{-1} B) Change in tropical cyclone (TC) frequency per intensity category for SSP245, 370, and 585 from the reference period 1995-2014 until 2100. C) Median relative sea level rise (RSLR) at mangrove locations for SSP245, 370, and 585 until 2100.*

Applying this methodology, we find widespread changes in tropical cyclone frequency across all levels of windspeed intensity and high rates of sea level rise, which put mangroves at varying levels of risk in all tropical cyclone basins (Figure 1). Even at the lowest emission pathway, many areas across the globe, including in Central America, Southeast Africa, East Asia and Oceania, will become newly affected by very high intensity tropical cyclone winds (Figure 1). Globally, depending on the SSP, we find 40-56% of mangrove area will be at high to severe risk (**Table 1**). A region of particular concern is Southeast Asia, where depending on the SSP, 52-78% of current mangrove extents are at high to severe risk (Figure 2, Supplementary Material Figure 5). In Southeast Africa, Central America, and Southwest Pacific, 48-86%, 51-71%, and 17-31% respectively are at high to severe risk depending on the SSP (**Figure 2**, Supplementary Material Figure 5). Hotspots of severe risk in Southeast Asia are in the Philippines, Vietnam, Malaysia, and parts of Indonesia (**Figure 1**, Panel A and **Figure 2**, Panel A). Areas in the Southwest Pacific, especially northern Australia, New Caledonia and Fiji, are also at severe risk (**Figure 1**, Panel A

and **Figure 2**, Panel A). In contrast, severe risk levels are less widespread in Central America and Southeast Africa but do occur at the east coast of Florida and the Bahamas, as well as in northern Mozambique and southern Tanzania (**Figure 1**, Panel A and **Figure 2**, Panel A).

Global mangrove ecosystem services and biodiversity at risk

To investigate risks to ecosystem service provision, we combine our risk index with the mangrove conservation priority areas by Dabalà et al. (2023), which rank global mangrove area based on mangrove species biodiversity, coastal protection for people and property, carbon sequestration and fishery benefits. We then investigate the distribution of risk levels for mangrove areas ranked as high conservation priorities areas (Dabalà et al., 2023), i.e. the top 10% and top 25% highest ranking areas in terms of their biodiversity, coastal protection for people and property, carbon sequestration and fishery benefits. Across all SSPs, these high priority mangrove areas are proportionally at higher risk than mangrove areas overall (**Table 1**). For example, under SSP370, 13% of global mangrove area will be at severe risk by the end of the century, while for the top ranking 25% it is 2% more, and for the top ranking 10% the proportion at severe risk is almost double than that of mangroves in general (**Table 1**).

Table 1. Percentages of current global mangrove extents at risk of tropical cyclone disturbance regime shift by 2100 for SSP245, SSP370, and SSP585. *In each cell the top number refers to the percentage of global mangrove extents, while the middle and bottom numbers refer to the proportion for the subsets of the top 25% and 10% ranked mangrove conservation priority areas, which are optimized based on mangrove species biodiversity, coastal protection for properties and people, carbon sequestration, and fishery benefits.*

Mangrove area [%] at risk of tropical cyclone disturbance regime shift globally

		None (0)	Low (1)	Medium (2)	High (3)	Very high (4)	Severe (5)
SSP245	Total	50.51	<0.01	9.24	24.35	14.62	1.27
	Top 25%	31.93	0.02	25.35	26.96	15.53	0.22
	Top 10%	29.07	0.02	11.40	34.69	24.00	0.82
SSP370	Total	52.70		1.30	9.60	23.51	12.88
	Top 25%	37.83	0	0.93	17.09	29.03	15.12
	Top 10%	30.20		0.02	11.36	33.44	24.98
SSP585	Total	44.44		<0.01	9.58	26.48	19.50
	Top 25%	28.57	0	0.02	20.25	28.45	22.72
	Top 10%	26.27		0.02	3.17	36.19	34.35

Zooming in on ecosystem service provision across regions, we observe distinct patterns of the areal extent and ecosystem services affected by varying degrees of risk (**Figure 2**). We project Southeast Asia to be most affected, both in terms of area and ecosystem services. Under SSP370, 98% and 97% of mangroves providing coastal protection to people and properties respectively are projected to be at high to severe risk, while 62% of carbon and 78% fishery benefits are highly to severely threatened under SSP 370 (**Figure 2**, Panel B). In Central America, especially coastal protection for assets is at severe risk, while in Southeast Africa about half of all services are at high to severe levels of risk. The Southwest Pacific is the least affected, with carbon sequestration being the most threatened service in the region.

Further, we observe heterogeneity in terms of proportions of services at risk compared to the locations' mangrove areal extent at risk. In all regions, the proportions of carbon sequestration and to a lesser degree also fishery benefits at given level of risk correspond closely to the proportion of overall mangrove area (**Figure 2**, Panel B). However, mangroves providing coastal

protection, both for people and property tend to be at higher levels of risk in all regions than mangrove area in general, except in Central America. This heterogenous distribution of some ecosystem services at risk in relation to mangrove area underscores the importance of mapping risk to services of interest explicitly.

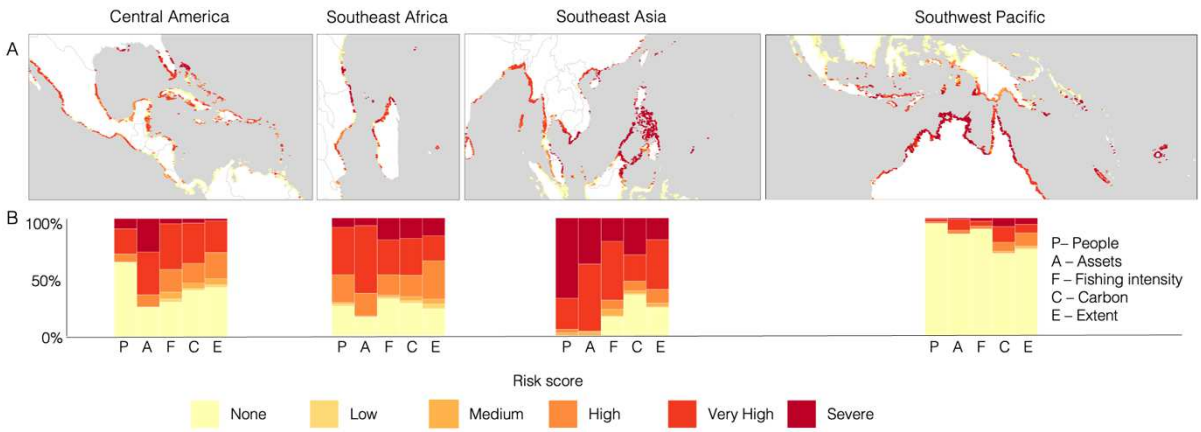

Figure 2: Risk hotspots and affected ecosystem services under SSP370 by 2100. *Panel A) shows the risk score for mangroves in the four hotspot areas. Panel B) shows the proportions of ecosystem services (people and assets protected, fishing intensity, and carbon storage), as well as the areal extent, at risk across the 5 risk scores low, medium, high, very high, and severe risk. Please see Supplementary Material Figures 4 and 5 for ecosystem services at risk under SSPs 245 and 585.*

Unchartered territory

Under climate change, disturbance regimes of different degrees of novelty will emerge (Turner & Seidl, 2023). We project that the frequency changes across different tropical cyclone windspeed intensities described above (**Figure 1**) will lead to some locations experiencing future disturbance regimes that resemble the current regimes of other areas, e.g. Central American mangroves receiving high intensity winds more frequently in the future, similarly to Southeast Asian mangroves currently. Other regions may become newly affected by tropical cyclones altogether, e.g. mangroves at higher latitudes due to a polewards shift in tropical cyclones. Hence, these projected regimes are locally novel but may be similar to historical regimes in other geographies.

We also project that some areas will experience a non-analog future, i.e. a *globally* novel disturbance regime, with tropical cyclones windspeed intensities at frequencies that are currently unprecedented. Figure 3 shows mangrove extents which are projected to experience novel tropical cyclone conditions under SSP 245, 370 and 585 by the end of the century, meaning that the projected future frequency exceeds the current global maximum frequency per given intensity category. Here, Southeast Asia and especially the Philippines clearly stand out, irrespective of intensity category or SSP. Historically, the global maximum frequency of tropical cyclone windspeeds above $70ms^{-1}$ is 0.02 (a return period of approximately 50 years). In the east of the Philippines, however, this frequency is projected to be exceeded by up to 254%, leading to a frequency of 0.08 (a return period of 12.5 years) (Figure 3).

Historically, the Philippines have been affected by a large share of all recorded tropical cyclone events and are exposed to storms that closely mirror the global distribution of tropical cyclones (Menéndez et al., 2020). The country also has total mangrove extents of more than 260,000 ha (approximately 2% of global extents) (Giri et al., 2011; Menéndez et al., 2020). Considering the frequency increases we find especially in high intensity tropical cyclone windspeeds in this area, it is uncertain in what state mangroves will be able to persist in the Philippines under unprecedented tropical cyclone frequencies across various levels of intensity. Given the relationship between tropical cyclone frequency and mangrove forest structural attributes, such as canopy height and aboveground biomass (Simard, Fatoyinbo, Smetanka, Rivera-Monroy, Castañeda-Moya, Thomas, & Van der Stocken, 2019), as well as community dynamics (Hogan et al., 2018), we may at least expect considerable structural changes under this unprecedented disturbance regime. Mangrove restoration may support recovery, however, restored mangrove forests are estimated to require 55 years to reach functional equivalency to their natural counterparts (Osland, Feher, Spivak, et al., 2020). Consequently, the nature and degree of future ecosystem service provision in these areas may also be subject to considerable change.

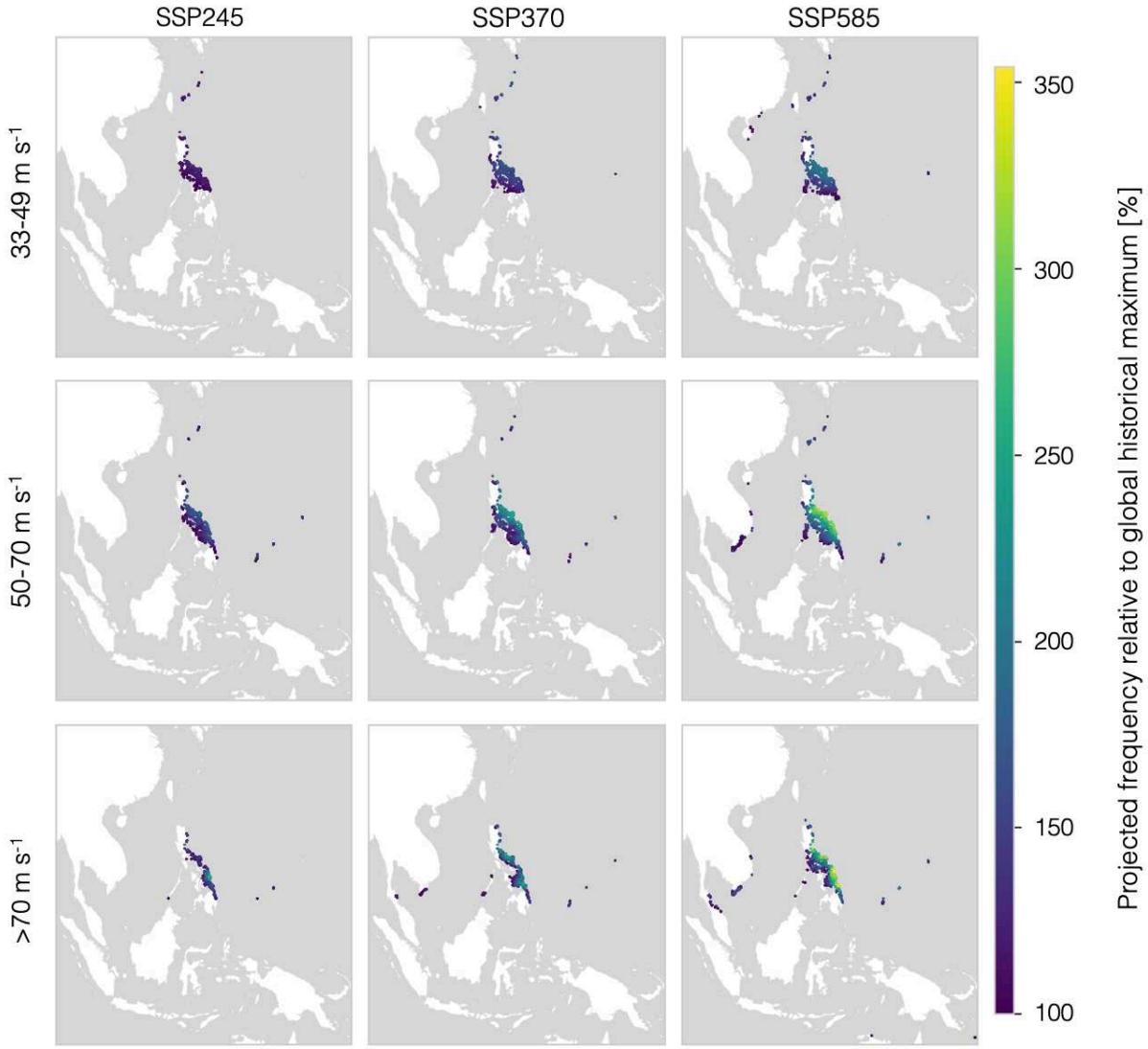

Figure 3: Mangrove extents affected by non-analogous tropical cyclone frequency by the end of the century per intensity category and SSP. *The colorbar indicates the projected frequency by 2100 as a percentage of the historical global maximum frequency per intensity category, with dark colors indicating projected frequencies of up to 200% of the historical global maximum frequency and lighter colors indicating projected frequencies up 350% of the historical global maximum. Only mangrove centroids with future TC frequencies higher than the historical global maximum per intensity category are shown. The historical maximum frequencies per category are 0.45 for TC1, 0.11 for TC2 and 0.02 for TC3, which corresponds to return periods of 2.22 years, 9.43 years, and 41.92 years respectively.*

Outlook and management implications

We present the first global, spatially explicit probabilistic projection of climate-modified changes in tropical cyclone disturbance regimes in combination with sea level rise affecting mangroves. This risk index incorporates changes in two stressors, tropical cyclones and sea level rise, and allows connecting the risk perspective with the spatial distribution of mangroves and the services they provide. Our analysis highlights widespread risk to mangrove areas and ecosystem services under climate change globally and identifies Southeast Asia as a region of especially high concern.

An improved understanding of the climate-induced changes in the tropical cyclone disturbance regime of mangrove ecosystems is vital in anticipating potential ecological responses of

mangrove forests to climate change. However, the findings presented here are subject to several uncertainties and limitations. First, the choice of tropical cyclone model, as well as the global climate model (GCM) data used to drive it influences the subsequent risk assessment (Meiler et al., 2022, 2023). For instance, using GCMs with high climate sensitivity has been shown to result in higher TC risk increases under climate change (Meiler et al., 2023). We account for this by using median TC frequency changes based on three GCMs of varying climate sensitivities in our results. Second, the choice of the relative change in frequency threshold beyond which to consider mangroves at risk is a modeling choice which affects the risk score assigned to the mangrove area and its associated ecosystem services. Our chosen threshold of a doubling in frequency by 2100 offers a balance between considering every deviation from the baseline novel and only considering the most extreme deviations as relevant (for more detail on the threshold choice, please consult the Methods). Third, a static distribution of mangrove extents and ecosystem services based on current conditions is assumed. Future research could integrate models for potential future mangrove distribution under other projected climatic changes, such as temperature and precipitation (Record et al., 2013) with the risk approach presented here.

To fully grasp novel ecological outcomes, it is not only necessary to analyze changes in the disturbance regime, but also investigate ecological responses and recovery mechanisms (Turner & Seidl, 2023). The global approach taken in this study does not allow including a variety of factors, which may greatly affect mangrove vulnerability and recovery such as mangrove forest structural properties, geomorphology, and anthropogenic disturbance (Amaral et al., 2023). While these factors could be included in studies aiming to mechanistically model the processes leading to ecological shifts, our approach is more suited to investigate characteristics of the changing disturbance regime and its spatial relation to mangroves and their services.

Our findings underline the necessity of adapting conservation and restoration planning to sustain ecosystem service provision under climate change and ecological transformation. We make progress towards this goal by mapping shifts in disturbance regimes, which has been identified as imperative for developing effective ecosystem management that is adaptive to changing disturbance regimes (Leverkus et al., 2021; Turner & Seidl, 2023). Considering the higher tropical cyclone frequency and associated severity of potential impacts, supporting ecosystem recovery through active restoration post-event may become necessary to support mangrove forest persistence and ecosystem service provision in some areas. Risk indices such as presented here could support restoration funding allocation, and contribute to accounting for losses of biodiversity and ecosystem services in Loss and Damage discussions (Roe et al., 2023). Further, our results point to the importance of considering future climate risk in the planning of blue carbon projects, especially to accurately quantify and mitigate reversal risks for nature-based blue carbon projects (Lovelock & Reef, 2020). In some cases, it may also unveil untapped potential for blue carbon by identifying areas not at risk of a shifting tropical cyclone regime, but where moderate rates of sea level rise could allow for high rates of vertical accretion and carbon capture (Saintilan et al., 2023).

Finally, this study underscores the valuable contributions of the hazard and risk modeling communities in studying emerging disturbance regimes under different levels of climate change and highlights opportunities for interdisciplinary research on the topic. Specifically, we point to

investigating further dimensions of disturbance novelty such as potential changes in seasonality or duration of disturbance. The risk framework presented here can be extended to include other climate-driven hazards relevant for mangrove ecosystems, such as drought and marine heatwaves, or applied to other sets of ecosystems and disturbances. Additionally, future research could integrate ecological responses and recovery through vulnerability functions calibrated based on vegetation characteristics and hazard intensity.

Our spatially explicit risk index, which combines climate change-driven changes in tropical cyclones with sea level rise outlines an emerging novel disturbance regime for mangroves under climate change by the end of the century. Further, our analysis highlights the importance of explicitly including ecosystem services in risk assessments, which are crucial to inform ecosystem-based climate mitigation and adaptation planning.

Methods

Tropical cyclone data

We use probabilistic tropical cyclone (TC) tracks from the MIT model (Emanuel, 2008; Emanuel et al., 2006), downscaled from 3 different CMIP6 models of varying climate sensitivity, namely ECEarth6, CESM2, and MIROC6, and under three Shared Socioeconomic Pathways (SSPs), SSP245, SSP370, and SSP585. TC tracks were generated for the historical period (1995-2014) and by the end of the century (2081-2100). Using the CLIMADA platform (Aznar-Siguan & Bresch, 2019), we generated two-dimensional wind fields by applying a parametric wind model (Holland, 2008) to each TC track at 150 arcseconds resolution. In detail, we compute the gridded 1-minute average sustained winds at 10 meters above ground. For each point in the wind field, we calculate both the historical (f_{hist}) and projected future (f_{fut}) frequency of windspeeds between 33-49 m s^{-1}, 50-70 m s^{-1}, and above 70 m s^{-1} respectively. The expected impacts for these wind intensities range from defoliation and occasional stem breakage to extensive canopy loss and finally widespread mortality and peat collapse (Krauss & Osland, 2020). Next, we calculate the absolute ($f_{fut} - f_{hist}$) and relative change in frequency ($(f_{fut} - f_{hist})/f_{hist}$) at the different windspeed intensities. The findings presented in the main text are based on median values of the three TC track sets based on the global climate models mentioned above. Visualizations of relative changes in TC frequency per climate model can be found in the Supplementary Material Figures 1-3.

Sea-level rise projections

As sea level rise data we use the medium-confidence regional sea-level rise projections from the IPCC AR6 ((Garner et al., 2021), https://doi.org/10.5281/zenodo.6382554), for SSP245, SSP370, and SSP585. Rates of sea level rise are reported for different levels of probability; we use the mean of medians across all IPCC AR6 workflows. We then calculate the median rate of regional relative sea level rise (RSLR) from 2020 until 2100 at each grid point (minimum 1 degree resolution, higher at coastline gauges). The median rates are then grouped into three intensity categories (0-4 mm y^{-1}, 4-7 mm y^{-1}, >7 mm y^{-1}), based on mangrove vertical adjustment capability reported by Saintilan et al. (2023). Vertical deficits are likely at RSLR rates beyond 4mm y^{-1}, and highly likely beyond 7 mm y^{-1} (Saintilan et al., 2023).

Mangrove data

We use the global mangrove extents polygons by the Global Mangrove Watch (GMW) (Bunting et al., 2018). The polygons are reprojected using an equal area projection (Eckert IV) to calculate their area and centroid coordinates. The resulting centroids are then matched with the tropical cyclone and regional relative sea-level rise data using the nearest-neighbor method.

Mangrove priority areas for conservation

Information on mangrove biodiversity and ecosystem services used was generated in a previous study by Dabalà et al. (2023) (available from https://doi.org/10.5281/zenodo.8272951). Dabalà et al. base their assessment on three different ecosystem services, namely coastal protection for people and assets, carbon storage, and fisheries benefits. The number of people and value of properties protected is based on a study by Menéndez et al. (2020). Carbon storage is estimated based on aboveground biomass (Simard, Fatoyinbo, Smetanka, Rivera-Monroy, Castañeda-

Moya, Thomas, & Van Der Stocken, 2019) and soil carbon (Sanderman et al., 2018). Fishing intensity data is based on zu Ermgassen et al. (zu Ermgassen et al., 2021). Dabalà et al. use a combination of mangrove biodiversity data which they intersect with the GMW global mangrove extents (Bunting et al., 2018), including mangrove species data from the IUCN Red List of Threatened species (IUCN, 2022), mangrove biophysical typology (T. A. Worthington et al., 2020), and marine ecoregions data (Spalding et al., 2007). Dabalà et al. have generated a variety of optimizations; we use the priority areas optimizing for biodiversity and ecosystem services (equally weighted) regardless of existing protected areas. The priority area planning units are hexagonal polygons at a resolution of 20km along the coastline based on the GMW mangrove distribution data (Dabalà et al., 2023). We assign the mangrove centroids used in the TC and RSLR risk analysis to the corresponding priority area hexagon using the nearest-neighbor method.

Mangrove TC & RSLR risk index

We compile the three intensity categories for TC and RSLR into a 3x3 risk matrix, assigning risk scores ranging from 1 to 5, where higher scores indicate greater risk. (yielding the risk categories low, medium, high, very high, and severe). For each mangrove centroid, we calculate the risk score based on the TC and RSLR category at the given location. Here, we only consider TC windspeeds with return periods of less than 250 years to be of ecological relevance (Kropf et al., 2024). The TC risk criterion is considered fulfilled if a centroid is newly affected by wind speeds of the given intensity or if the frequency is projected to at least double $((f_{fut} - f_{hist})/f_{hist} \geq 1)$. If this criterion is met across multiple TC intensity categories at the same centroid, the highest possible risk rank is assigned. Since we focus primarily on the shift in TC risk, we do not include areas only affected by sea level rise in the risk index.

The relative change in frequency threshold beyond which to consider a centroid at risk is modeling choice, which affects the final risk classification, and has implications for the mangrove area and ecosystem services reported to be at risk. Novelty of a disturbance regime emerges across a spectrum and can be at varying degrees of departure from the historical range of variability (Turner & Seidl, 2023). One could consider any deviation from the baseline frequency novel; however, we take a more conservative approach in setting the threshold at a 100% increase, i.e. doubling, beyond the historical baseline. Please see SM Table 1 for an overview of the share of mangrove centroids above a 50%, 100%, and 200% increase in frequency. We chose the 100% threshold, since a doubling in frequency represents a clear deviation from the baseline, while also maintaining variation in terms of the share of mangrove centroids meeting this criterium across different intensity categories and SSPs.

Ecosystem service risk analysis

The results presented in **Table 1** are based on the priority rank assigned to each individual mangrove centroid and the corresponding risk rank per centroid. The matching between risk score and priority unit is achieved by assigning the mangrove centroids containing the risk score information with the priority area planning units using the nearest-neighbor method, as described above. Here, we compare the risk index of the top-ranked 10% and 25% (i.e. 10th and 25th percentile priority rank, which corresponds to a rank of equal to or higher than 2 or 8 respectively) mangrove centroids to the risk index of all mangrove centroids globally. Since Dabalà et al. report the absolute values of ecosystem services provided by mangroves per hexagonal planning unit,

the proportions of ecosystem services at various levels of risk shown in Panel B of **Figure 2** are also calculated at the level of the planning units. To obtain the risk index at the scale of the planning units rather than the individual centroids, we calculate an area-weighted mean risk index based on the areal extent and risk index values of all mangrove centroids contained in each planning unit. Despite some loss in the resolution of the risk index, we follow this approach to avoid misrepresenting the total amount of ecosystem services provided in a given area. The area-weighted risk score $R_{weighted}$ per priority area planning unit P is calculated by dividing the sum of the products of each risk score r of mangrove centroids contained in the given planning unit and its associated area A with the total area of the given planning unit (Equation 1).

$$R_{weighted}(P) = \frac{\sum_r \left(r \times A(P,r)\right)}{\sum_r A(P,r)}$$

Equation 1

This spatial overlap approach assumes a static provision of ecosystem services and does not take into account potential human responses to changes in extreme events or fluctuations in ecosystem service provision. However, it does allow identifying where areas with currently high ecosystem service provision coincide with future risk under climate change.

Non-analogous tropical cyclone frequencies

The mangroves affected by non-analogous tropical cyclone frequencies under SSP245, 370, and 585 (Figure 3) are calculated as follows: First, we filter for all centroids where the projected future frequency (f_{fut}) exceeds the *global* historical maximum frequency (f_{max}) within each windspeed intensity category. Next, we calculate the change ratio ($C_{nonanalog}$) between the projected future frequency (f_{fut}) for the given mangrove centroid and the global historical maximum frequency (f_{max}) per windspeed intensity category:

$$C_{nonanalog} = \frac{(f_{fut} - f_{max})}{f_{max}}$$

Equation 2

Author contributions

Conceptualization by S.H., L.E.D., C.M.K, S.M., and D.N.B; Methodology by S.H.; Formal analysis by S.H.; Data curation: S.H., S.M.; Writing – original draft by S.H.; Writing – review & editing by S.H., L.E.D., C.M.K, S.M., and D.N.B. Funding procurement – D.N.B.

Competing Interests

The authors declare no competing interests.

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